Discrete-Time Markov Chains (DTMCs): Part 1

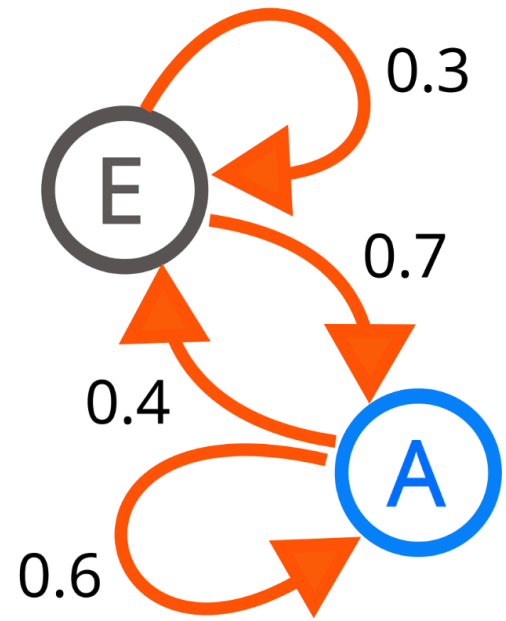
Gaurav S. Kasbekar
Dept. of Electrical Engineering
IIT Bombay

References

- S.M. Ross, “Stochastic Processes”: Chapter 4
- R. Wolff: Chapter 3
- S.M. Ross, “Introduction to Probability Models”: Chapter 4
- M. Harchol-Balter: Chapter 8

Discrete-Time Markov Chains

- Consider a stochastic process $\{X_n, n = 0,1,2, \dots\}$ that takes values from a countable state space S
- Unless otherwise mentioned, we assume that $S = \{0,1,2, \dots\}$
- If $X_n = i$, then process said to be in *state* i at time n
- $\{X_n\}$ said to be a discrete-time Markov chain (DTMC) if:
  - ❑ $P(X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, \dots, X_1 = i_1, X_0 = i_0) = P(X_{n+1} = j | X_n = i)$ for all states $i_0, i_1, \dots, i_{n-1}, i, j$, and all $n \geq 0$
- E.g.:
  - ❑ $P(X_4 = 10 | X_3 = 5, X_2 = 12, X_1 = 3, X_0 = 7) = P(X_4 = 10 | X_3 = 5)$
- *Markovian Property*:
  - ❑ Given present state X_n, future state X_{n+1} is independent of past states $X_0, X_1, \dots, X_{n-1}$
- For $i, j \in \{0,1,2, \dots\}$, let $P_{ij} = P(X_{n+1} = j | X_n = i)$:
  - ❑ probability that when process is in state i, it will make a transition into state j
  - ❑ called *transition probability*



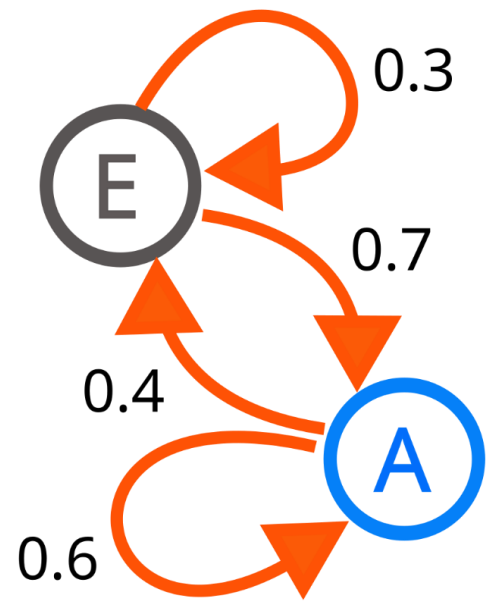
Ref: https://en.wikipedia.org/wiki/Markov_chain

Transition Probabilities

- Recall:
  - ❑ transition probabilities are $P_{ij} = P(X_{n+1} = j | X_n = i)$, where $i, j \in \{0,1,2,\dots\}$
- Properties:

  1) $P_{ij} \geq 0$, for all $i, j \in \{0,1,2,\dots\}$

  2) $\sum_{j=0}^{\infty} P_{ij} = 1$, for all $i \in \{0,1,2,\dots\}$

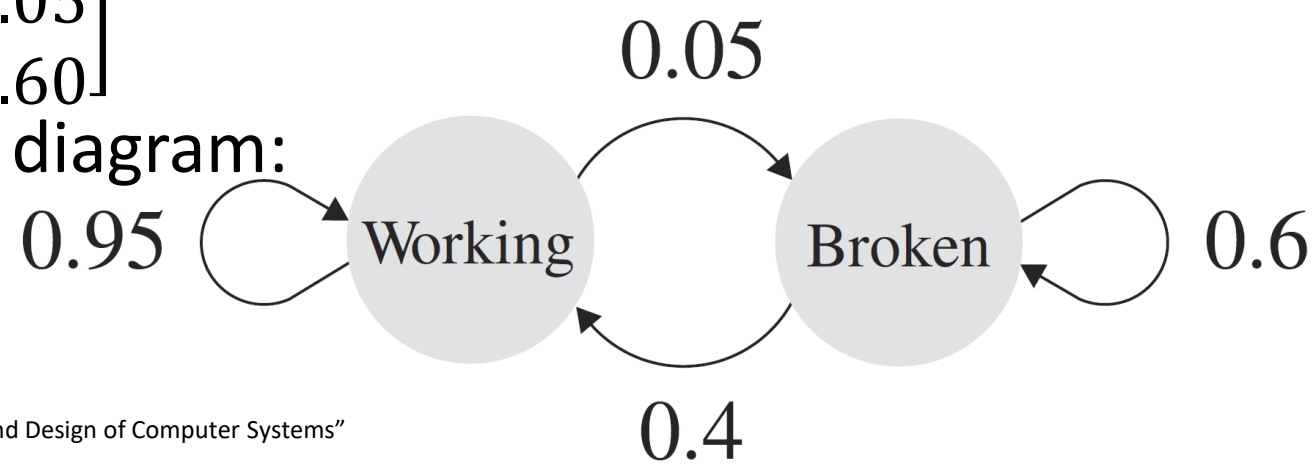


- Let P be matrix of transition probabilities:
  - ❑ $P = \begin{bmatrix} P_{00} & P_{01} & P_{02} & \cdots \\ P_{10} & P_{11} & P_{12} & \cdots \\ \vdots & & & \\ P_{i0} & P_{i1} & P_{i2} & \cdots \\ \vdots & & & \vdots \end{bmatrix}$
- If there are an infinite number of states in S, then P has infinite order
- P called *stochastic matrix*:
  - ❑ which is a matrix that satisfies 1) and 2)

Ref: https://en.wikipedia.org/wiki/Markov_chain

Example 1: Repair Facility Problem

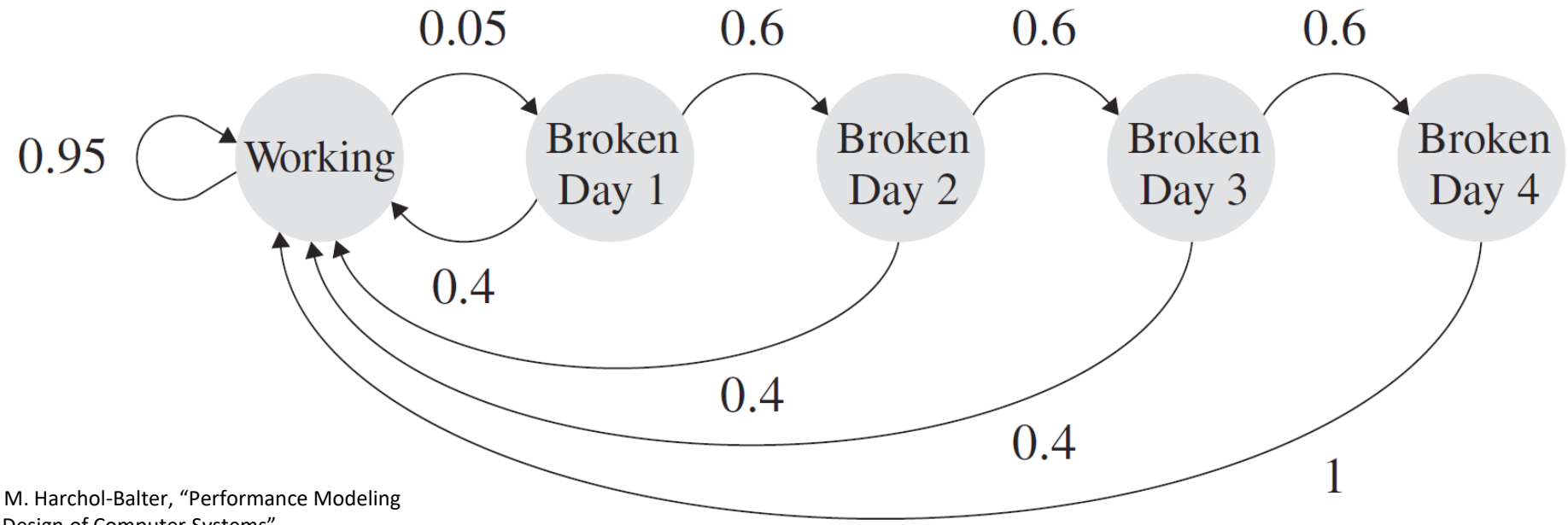
- A machine is either working or in repair center
- If it is working today, then there is a $95\ \%$ chance that it will be working tomorrow
- If it is in repair center today, then there is a $40\ \%$ chance that it will be working tomorrow
- States of DTMC:
  - 0: Working
  - 1: Broken (In repair center)
- Transition probability matrix:
  - $P = \begin{bmatrix} 0.95 & 0.05 \\ 0.40 & 0.60 \end{bmatrix}$
- State transition diagram:
  - shown in fig.



Ref: M. Harchol-Balter, "Performance Modeling and Design of Computer Systems"

Example 1: Repair Facility Problem (contd.)

- Now suppose that after machine remains broken for four days:
  - It is replaced with a new machine
- State transition diagram:
  - shown in fig.



Ref: M. Harchol-Balter, "Performance Modeling and Design of Computer Systems"

Example 2: The M/G/1 Queue

- Consider an M/G/1 queue with service distribution $G(\cdot)$
- Let $X(t)$ be number of customers in system at time t
- Does $\{X(t): t \geq 0\}$ have Markovian property?
  - ❑No, since future evolution of system depends on how long the customer in service has already been there
- Consider the stochastic process $X_1, X_2, \ldots$ obtained by looking at above system at instants when customers depart:
  - ❑X_n: no. of customers left behind by nth departure, for $n \geq 1$
- For $n \geq 1$, let Y_n: number of customers arriving during service period of $(n+1)$st customer
- X_{n+1} in terms of X_n:

1) $$\begin{cases} X_n - 1 + Y_n, & \text{if } X_n > 0, \\ Y_n, & \text{else.} \end{cases}$$

Example 2: The M/G/1 Queue (contd.)

- Recall that X_{n+1}:

  1) $\begin{cases} X_n - 1 + Y_n, & \text{if } X_n > 0, \\ Y_n, & \text{else.} \end{cases}$

- Distribution of Y_n, $n \geq 1$:

  2) $P\{Y_n = j\} = \int_0^\infty e^{-\lambda x} \frac{(\lambda x)^j}{j!} dG(x)$, for $j \in \{0,1,2,\dots\}$

  - also, Y_n, $n \geq 1$, are independent

- $\{X_n, n = 1,2,\dots\}$ is a DTMC with following transition probabilities:

  - $P_{0j} = \int_0^\infty e^{-\lambda x} \frac{(\lambda x)^j}{j!} dG(x)$, for $j \in \{0,1,2,\dots\}$
  - $P_{ij} = \begin{cases} \int_0^\infty e^{-\lambda x} \frac{(\lambda x)^{j-i+1}}{(j-i+1)!} dG(x), & \text{if } j \geq i-1, i \geq 1, \\ 0, & \text{else.} \end{cases}$

- Above DTMC is obtained by looking at original process only at certain time instants:

  - known as *embedded DTMC*

Example 3: The G/M/1 Queue

- Consider a G/M/1 queue such that customers arrive in accordance with a renewal process with interarrival distribution $G(\cdot)$
- Suppose service distribution is exponential with rate μ
- Let $X(t)$ be number of customers in system at time t
- Does $\{X(t): t \geq 0\}$ have Markovian property?
  - ❑No, since future evolution of system depends on amount of time that has elapsed since a customer arrived
- Consider the stochastic process $X_1, X_2, \ldots$ obtained by looking at above system at instants when customers arrive:
  - ❑X_n: number of customers in system as seen by nth arrival, for $n \geq 1$
- $\{X_n, n = 1,2, \ldots\}$ is a DTMC

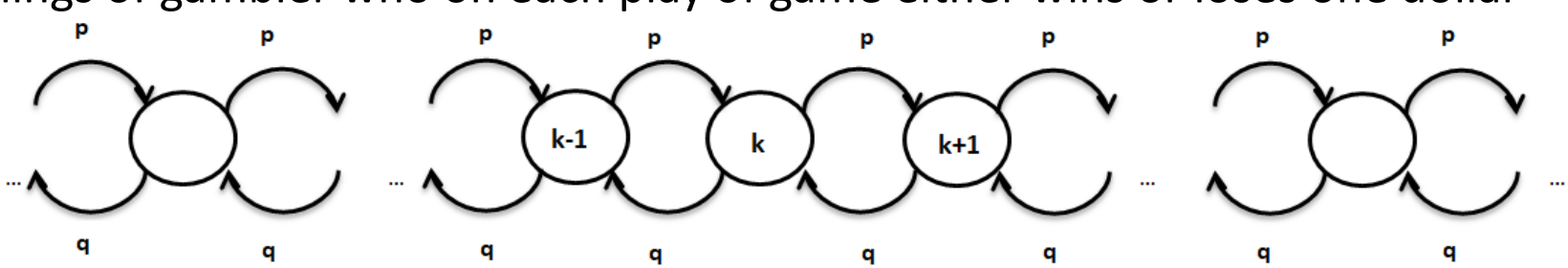
Example 3: The G/M/1 Queue (contd.)

- Distribution of number of customers served in an interval of length t, assuming that there are some customers to be served throughout the interval:
  - Poisson with mean μt
- $P_{i,i+1-j}$:
  - $\int_0^\infty e^{-\mu t} \frac{(\mu t)^j}{j!} dG(t)$, for $j \in \{0,1,\dots,i\}$
- $P_{i,0}$:
  - probability that at least $i+1$ Poisson events occur in random length of time having distribution G
  - $\int_0^\infty \left(\sum_{k=i+1}^\infty e^{-\mu t} \frac{(\mu t)^k}{k!} \right) dG(t)$, for $i \in \{0,1,\dots\}$
- Above DTMC is another example of an embedded DTMC

Example 4: Random Walk

- Let X_i, $i \geq 1$, be i.i.d. with:
  - $P\{X_i = j\} = a_j$, for $j \in \{0, \pm 1, \dots\}$
- Let:
  - $S_0 = 0$,
  - $S_n = \sum_{i=1}^{n} X_i$
- Then $\{S_n, n = 0,1,2, \dots\}$:
  - is a DTMC with transition probabilities $P_{ij} = a_{j-i}$
  - called a *general random walk*
- Following special case called *simple random walk*:
  - $a_j = \begin{cases} p, & \text{if } j = 1, \\ q, & \text{if } j = -1, \\ 0, & \text{else.} \end{cases}$
  - where $0 < p < 1$ and $q = 1 - p$
- Interpretation:
  - wanderings of drunken person as he/ she walks along straight line, or
  - winnings of gambler who on each play of game either wins or loses one dollar

Ref: https://www.analyticsvidhya.com/blog/2021/02/markov-chain-mathematical-formulation-intuitive-explanation-applications/



Chapman-Kolmogorov Equations

- For $i, j \geq 0$ and $n \geq 0$, let:
  - ❑ $P_{ij}^{n} = P\{X_{n+m} = j | X_m = i\}$,
  - ❑ called n-step transition probabilities
- Note that $P_{ij}^{1} = P_{ij}$ for all $i, j \geq 0$
- Suppose we know P_{ij}^{n} and P_{ij}^{m} for some $m, n \geq 0$ and all $i, j \geq 0$
- Then P_{ij}^{n+m}:
  - ❑ $P\{X_{n+m} = j | X_0 = i\}$
  - ❑ $\sum_{k=0}^{\infty} P\{X_{n+m} = j | X_n = k, X_0 = i\} P\{X_n = k | X_0 = i\}$
  - ❑ $\sum_{k=0}^{\infty} P_{kj}^{m} P_{ik}^{n}$
- Hence:

  1) $P_{ij}^{n+m} = \sum_{k=0}^{\infty} P_{ik}^{n} P_{kj}^{m}$, for all $m, n \geq 0$ and all $i, j \geq 0$
- Equations 1) known as *Chapman-Kolmogorov equations*
- Let $P^{(n)}$ be matrix of n-step transition probabilities P_{ij}^{n}
- Then $P^{(n+m)}$:
  - ❑ $P^{(n)} P^{(m)}$
- Hence, for each $n \geq 1$, $P^{(n)}$:
  - ❑ P^{n}

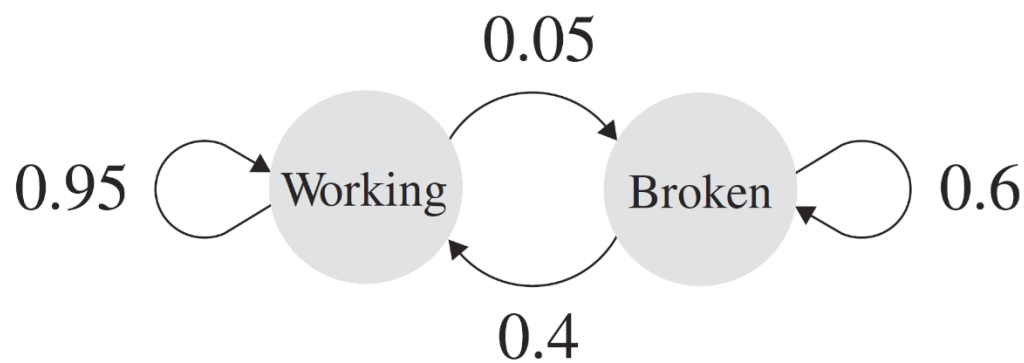
Example: Repair Facility Problem (contd.)

- States of DTMC:
  - ❑ 0: Working
  - ❑ 1: Broken (In repair center)
- General transition probability matrix:
  - ❑ $P = \begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix}$,
  - ❑ where $0 < a < 1$ and $0 < b < 1$
- P^n:

$$1) \quad \begin{bmatrix} \frac{b+a(1-a-b)^n}{a+b} & \frac{a-a(1-a-b)^n}{a+b} \\ \frac{b-b(1-a-b)^n}{a+b} & \frac{a+b(1-a-b)^n}{a+b} \end{bmatrix}$$

  - ❑ **Exercise**: Prove 1) by induction
- $\lim_{n\to\infty} P^n$:

$$2) \quad \begin{bmatrix} \frac{b}{a+b} & \frac{a}{a+b} \\ \frac{b}{a+b} & \frac{a}{a+b} \end{bmatrix}$$



- Note that all rows in 2) are the same
- Later we find general conditions under which this is true for a DTMC

Ref: M. Harchol-Balter, "Performance Modeling and Design of Computer Systems"

Communication

- State j is said to be accessible from state i if for some $n \geq 1$, $P_{ij}^{n} > 0$:
  - alternatively, we say state i can reach state j
  - denoted by $i \to j$
- $i \to j$ iff:
  - there is a path (finite sequence of transitions) of positive probability that starts in i and ends in j
- Two states i and j accessible to each other are said to *communicate* and we write:
  - $i \leftrightarrow j$
- By definition, we say that for every state i:
  - $i \leftrightarrow i$
  - i.e., every state communicates with itself
- **Proposition**: Communication is an equivalence relation, i.e., for states i, j, k:

  i. $i \leftrightarrow i$;

  ii. If $i \leftrightarrow j$, then $j \leftrightarrow i$;

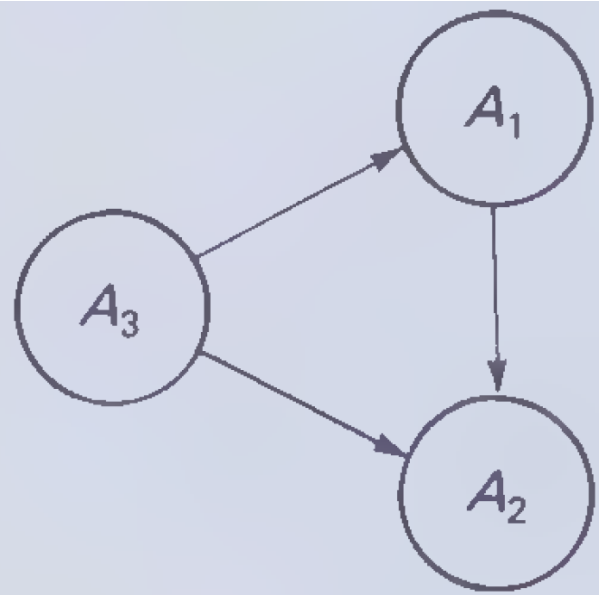
  iii. If $i \leftrightarrow j$ and $j \leftrightarrow k$, then $i \leftrightarrow k$.
- **Proof**:
  - First two parts trivially follow from definition
  - To prove iii., suppose there exist m and n such that $P_{ij}^{m} > 0$ and $P_{jk}^{n} > 0$
  - Then P_{ik}^{m+n}:
    - $\sum_{r=0}^{\infty} P_{ir}^{m} P_{rk}^{n} \geq P_{ij}^{m} P_{jk}^{n} > 0$

Communication (contd.)

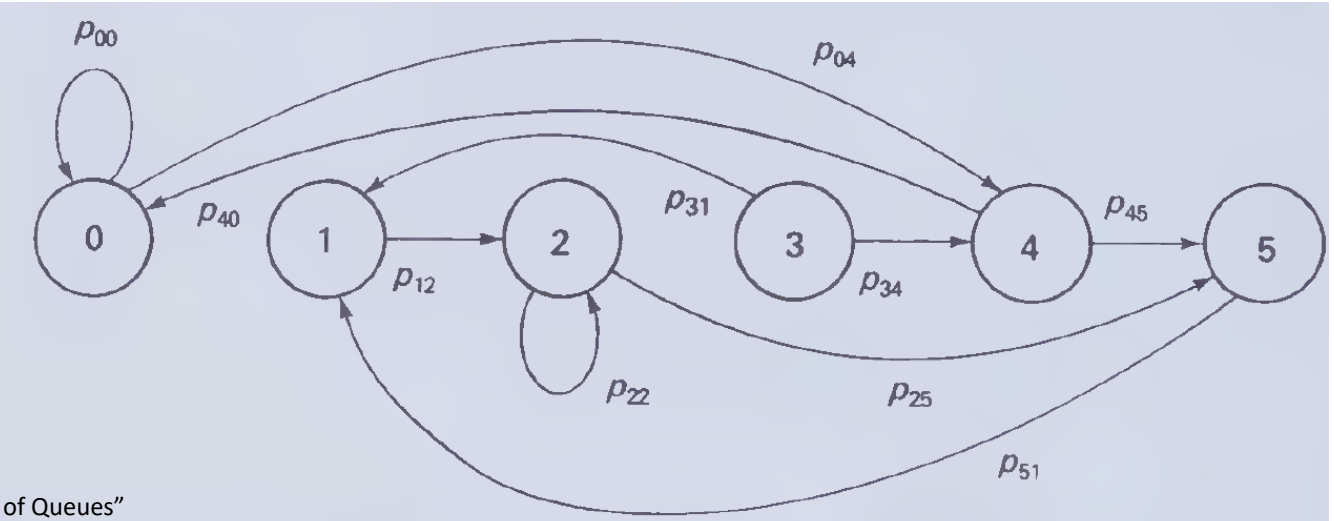
- Recall:
  - **Proposition**: Communication is an equivalence relation, i.e., for states i, j, k:
    - *i.* $i \leftrightarrow i$;
    - ii. If $i \leftrightarrow j$, then $j \leftrightarrow i$;
    - iii. If $i \leftrightarrow j$ and $j \leftrightarrow k$, then $i \leftrightarrow k$.
- Two states that communicate are said to be in same *communication class*
- Any two classes are either disjoint or identical:
  - this follows from above Proposition
- We can partition the set of states S of the DTMC into communication classes $S_1, S_2, \ldots$ with the following properties:
  - a) Every state i belongs to exactly one class, say S_k,
  - b) Every class S_k contains at least one state,
  - c) States in the same class communicate with each other,
  - d) States belonging to different classes do not communicate
- The DTMC is said to be *irreducible* if:
  - there is only one class, i.e., all states communicate with each other

Example

- Fig. below shows:
  - a transition probability matrix, where $(*)$ denotes the position of each positive element, and
  - state transition diagram corresponding to this matrix
- What are the communication classes of this DTMC?
- $0 \leftrightarrow 4$:
  - since $0 \to 4$ and $4 \to 0$
- Also, $5 \leftrightarrow 1 \leftrightarrow 2$:
  - since $5 \to 1 \to 2 \to 5$
- No state can reach state 3
- No state in $\{5,1,2\}$ can reach $\{0,4\}$
- Hence, the communication classes are:
  - $A_1 = \{0,4\}$
  - $A_2 = \{5,1,2\}$
  - $A_3 = \{3\}$
- "Can reach" relations between different classes:
  - are shown in fig. on right



$$P = \begin{pmatrix} * & 0 & 0 & 0 & * & 0 \\ 0 & 0 & * & 0 & 0 & 0 \\ 0 & 0 & * & 0 & 0 & * \\ 0 & * & 0 & 0 & * & 0 \\ * & 0 & 0 & 0 & 0 & * \\ 0 & * & 0 & 0 & 0 & 0 \end{pmatrix}$$



Ref: R. Wolff, "Stochastic Modeling and the Theory of Queues"

Recurrence and Transience

- For any states i and j, let:
  - $f_{ij}^{n} = P\{X_n = j, X_k \neq j, k = 1, \dots, n-1 | X_0 = i\}$ for $n \geq 1$
  - probability that, starting in i, first transition into j occurs at time n
- As a convention, we let $f_{ij}^{0} = 0$ for all i and j
- Probability of ever making a transition into state j, given that process starts in i:
  - $f_{ij} = \sum_{n=1}^{\infty} f_{ij}^{n}$
- For $i \neq j$, $f_{ij} > 0$ iff:
  - j is accessible from i
- State j is said to be:
  - *recurrent* if $f_{jj} = 1$
  - *transient* if $f_{jj} < 1$

A Necessary and Sufficient Condition for Recurrence

- **Proposition**:
  - ❑ State j is recurrent iff:
    - $\sum_{n=1}^{\infty} P_{jj}^{n} = \infty$
- **Proof**:
  - ❑ State j is recurrent if, w.p. 1, process starting at j will eventually return
  - ❑ By Markovian property, process probabilistically restarts itself upon returning to j
  - ❑ Hence, if j is recurrent and $X_0 = j$, then the number of visits to j:
    - will be infinite w.p. 1 and hence will have mean ∞
  - ❑ If j is transient and $X_0 = j$, then the number of visits to j:
    - will be geometric with finite mean $\frac{f_{jj}}{1-f_{jj}}$
  - ❑ Thus, state j is recurrent iff:
    - $E(\text{number of visits to } j | X_0 = j) = \infty$
  - ❑ Now, $E(\text{number of visits to } j | X_0 = j)$:
    - $E\left(\sum_{n=1}^{\infty} 1_{\{X_n = j\}} \middle| X_0 = j\right)$
    - $\sum_{n=1}^{\infty} E\left(1_{\{X_n = j\}} \middle| X_0 = j\right)$
    - $\sum_{n=1}^{\infty} P_{jj}^{n}$
- Above proof also shows that w.p. 1, a transient state will only be visited a finite number of times:
  - ❑ hence the name “transient”

Recurrence and Transience Are Class Properties

- **Lemma**:
  - If i is recurrent and $i \leftrightarrow j$, then j is recurrent
- **Proof**:
  - Let m and n be such that $P_{ij}^{n} > 0$ and $P_{ji}^{m} > 0$
  - For any $s \geq 0$:
    - $P_{jj}^{m+n+s} \geq P_{ji}^{m} P_{ii}^{s} P_{ij}^{n}$
  - So, $\sum_{s=1}^{\infty} P_{jj}^{m+n+s} \geq$:
    - $P_{ji}^{m} P_{ij}^{n} \sum_{s=1}^{\infty} P_{ii}^{s} = \infty$
  - Hence, result follows from above Proposition
- Above lemma implies following result
- **Theorem**: The states in a communication class A are either:
  - all recurrent, or
  - all transient

Example

- Recall simple random walk:
  - ❑ State space is set of all integers $\{0, \pm 1, \dots\}$
  - ❑ Transition probabilities $P_{i,i+1} = p$ and $P_{i,i-1} = 1 - p$ for all $i \in \{0, \pm 1, \dots\}$:
    - where $0 < p < 1$
- Want to find out which states are recurrent and which transient
- Since all states communicate, it follows from above Proposition that either all states are transient or all recurrent
- Consider state 0 and let us find:
  - ❑ whether $\sum_{n=1}^{\infty} P_{00}^{n}$ is finite or infinite
- Since starting from 0, process cannot return to 0 after odd number of steps:
  - ❑ $P_{00}^{2n+1} = 0$, for all $n \in \{1,2, \dots\}$
- Also, for each $n \in \{1,2, \dots\}$, P_{00}^{2n}:
  - ❑ $\binom{2n}{n} p^n (1-p)^n$
  - ❑ $\frac{(2n)!}{n!n!} \big(p(1-p)\big)^n$

Example (contd.)

- Recall: for each $n \in \{1,2,\dots\}$, P_{00}^{2n}:
  - $\frac{(2n)!}{n!n!}\left(p(1-p)\right)^n$
- By Stirling's approximation:
  - $n! \sim \left(n^{n+\frac{1}{2}}\right)e^{-n}\sqrt{2\pi}$
  - where $a_n \sim b_n$ if $\lim_{n\to\infty} \frac{a_n}{b_n} = 1$
- So, P_{00}^{2n}:
  - $\sim \frac{\left(4p(1-p)\right)^n}{\sqrt{\pi n}}$
- **Exercise**: Show that if $a_n \sim b_n$, then $\sum_{n=1}^{\infty} a_n < \infty$ iff $\sum_{n=1}^{\infty} b_n < \infty$
- Hence, $\sum_{n=1}^{\infty} P_{00}^{n} < \infty$ iff following quantity is finite:
  - $\sum_{n=1}^{\infty} \frac{\left(4p(1-p)\right)^n}{\sqrt{\pi n}}$
- Now, $4p(1-p) \leq 1$, with equality holding iff $p = \frac{1}{2}$
- So, $\sum_{n=1}^{\infty} P_{00}^{n} = \infty$ iff $p = \frac{1}{2}$
- Thus, the DTMC is:
  - recurrent if $p = \frac{1}{2}$
  - transient if $p \neq \frac{1}{2}$

Closed and Open Sets

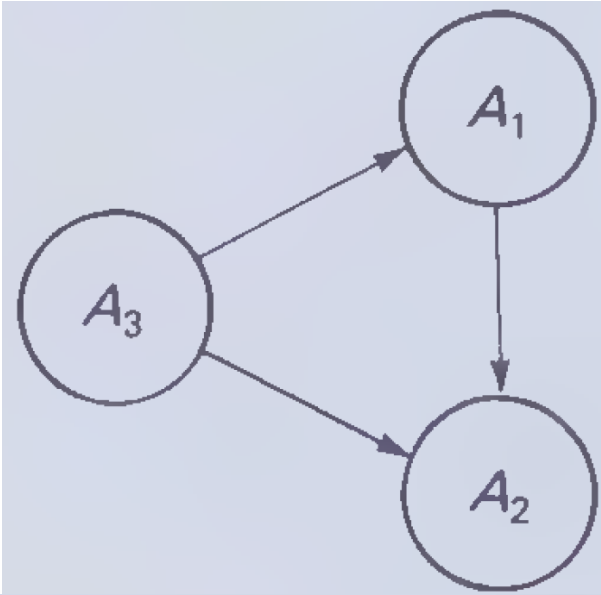
- A set of states C is said to be *closed* if:
  - For every pair of states $i \in C$ and $j \notin C$, $P_{ij} = 0$
- Clearly:
  - Paths of positive probability starting in C must remain in C
  - $P_{ij}^{(n)} = 0$ for every $n \geq 1$
- If C is not closed, it is said to be *open*

Transience of Open Communication Classes

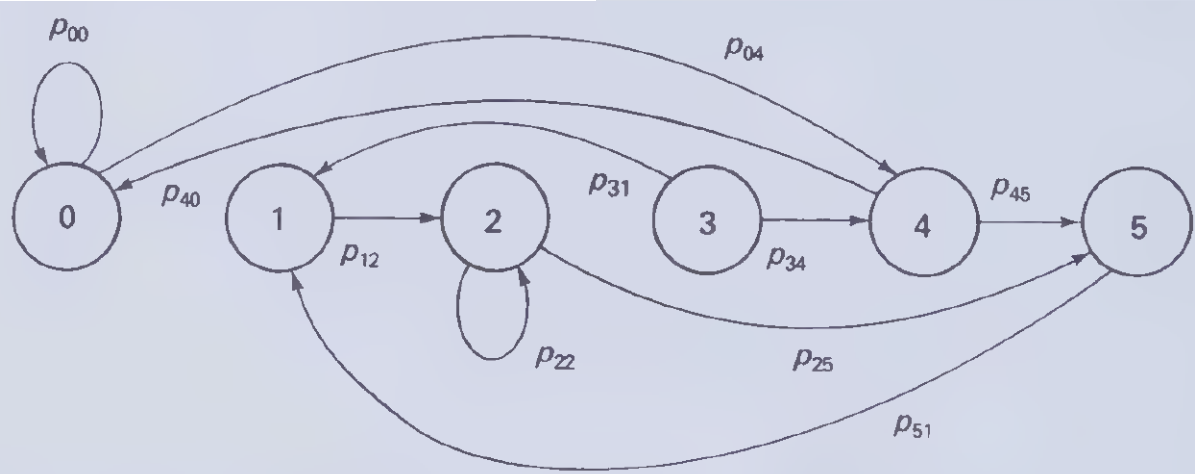
- **Theorem**: Open communication classes are transient
- **Proof**:
  - ❑Let A be an open communication class
  - ❑So, there exist states $i \in A$ and $j \notin A$ s.t. $P_{ij} > 0$
  - ❑Hence, $i \to j$, but $j \nrightarrow i$:
    - ○ otherwise j would be in A
  - ❑So, $f_{ji} = 0$
  - ❑Suppose $X_0 = i$
  - ❑Conditioning on whether the next transition is to state j:
    - ○ $f_{ii} = f_{ji}P_{ij} + P(\text{process returns to state } i | X_1 \neq j)\left(1 - P_{ij}\right)$
    - ○ $\leq 0 + \left(1 - P_{ij}\right) < 1$
  - ❑So, state i is transient

Recall: Example

- The communication classes are:
  - ❑ $A_1 = \{0,4\}$
  - ❑ $A_2 = \{5,1,2\}$
  - ❑ $A_3 = \{3\}$
- "Can reach" relations between different classes are shown in fig. on right
- Which classes are transient and which are recurrent?
- Classes A_1 and A_3 are open and hence transient
- Class A_2 is closed and finite; later, we will show that this implies it is recurrent



$$P = \begin{pmatrix} * & 0 & 0 & 0 & * & 0 \\ 0 & 0 & * & 0 & 0 & 0 \\ 0 & 0 & * & 0 & 0 & * \\ 0 & * & 0 & 0 & * & 0 \\ * & 0 & 0 & 0 & 0 & * \\ 0 & * & 0 & 0 & 0 & 0 \end{pmatrix}$$



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